Name: _____ **Marks:** _____/32

1. **Outline the key differences between data and information? (2 marks)**
2. **Give 2 primary purposes of a database. (4 marks)**
3. **An organization needs to improve its current computer systems. The systems are legacy systems with a large number of end users.**

   a) Identify two issues concerning the roles of end users that must be considered in relation to the new system. (2 marks)

   b) Outline the meaning of the term “legacy system”. (2 marks)

   c) Identify two methods of gathering requirements from end users. (2 marks)

   The organization needs to use existing data in the new system.

   d) Explain one problem that may occur during data migration. (3 marks)

   A decision needs to be made on whether to use phase conversion or a direct changeover method of implementation.

   e) Give one advantage and disadvantage of each method of implementation. (4 marks)

4. **Identify and write the output that will be displayed in the given code. (2 marks)**

```
def add(a, b):
    return a+5, b+5

result = add(3, 2)
print(result)
```

5. **Identify and write the output that will be displayed in the given code. (2 marks)**

```
1  total = 0
2  for i in range(1, 6):
3      total += i
4  print(f"The sum of numbers from 1 to 5 is {total}")
```

6. **Galaxy Bank is a US based bank with many banks (branches) across the country. Galaxy Bank uses a relational database to support its operations. (9 marks)**

ACCOUNTS

Acct_no	Customer_ID	Balance	Account_Type
171345090	987887982	212456.00	Deposit
171345087	987887982	3450.00	Current
345904570	456783356	320567.00	Deposit
345904571	456783356	590.00	Current
435362728	987635218	610.00	Current
356278236	533763373	4567.00	Current
...	...	...	...

CUSTOMERS

Customer_ID	Family_Name	Zip_Code	Phone
668848616	Smith	90906	3456042
987887982	Wallace	32001	3205600
124567552	James	23057	6783456
648782904	Evans	21026	4545542
456783356	Donnan	23058	3789036
453627286	Davies	24026	3678936
546373883	Walker	25098	3974673
...	...	...	...

a) With reference to the tables ACCOUNTS and CUSTOMERS, write an SQL query to find the names of customers whose account balance is greater than $300 000. (5 marks)

b) State the result of the following query. (4 marks)

```
SELECT a.acct_no, c.family_name, a.balance, c.phone
FROM ACCOUNTS a
JOIN CUSTOMERS c ON a.Customer_ID = c.Customer_ID
WHERE a.Balance < 1000;
```